

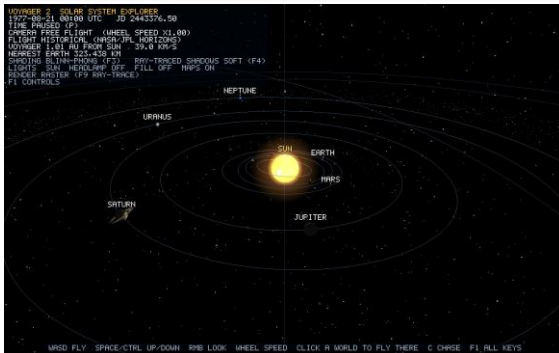
Voyager 2 Solar-System Explorer

01 INTRODUCTION · What the project is and why it was built

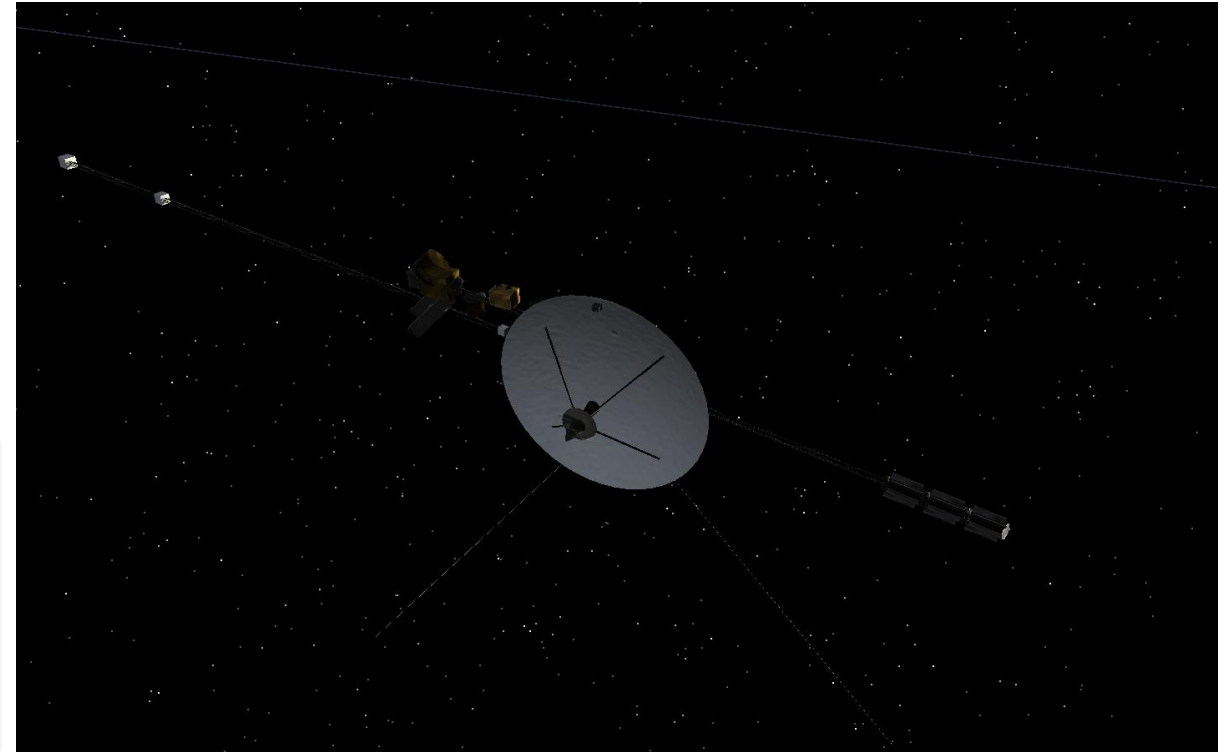
Explore a 3D solar system.
Explain how it is rendered.

Replay flybys and inspect the spacecraft.
Fly manually. Compare rendering techniques.

26 textured bodies · 4 ring systems
76 assemblies · 18 hardware targets
8,500 instanced rocks · dynamic comet



C++20 · OpenGL 3.3 · GLFW / GLAD / GLM

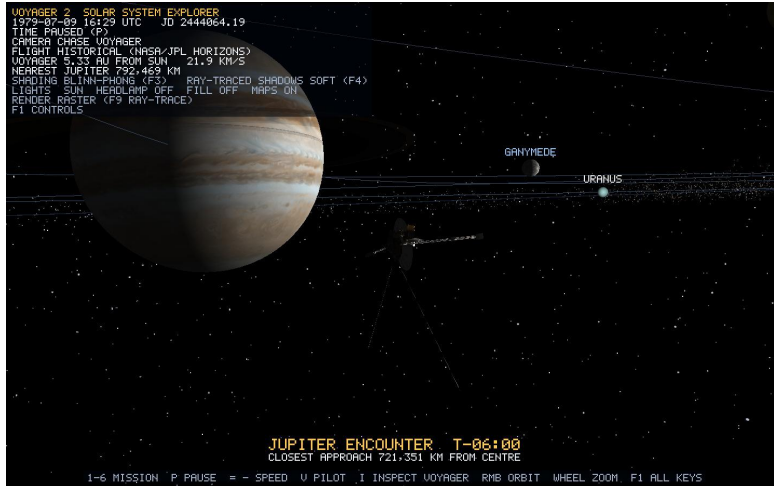


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CSE 4102 | KUET | October 2026

Course teachers: **Md Tajmilur Rahman** - Lecturer
and **Md Mubtashim Abrar Nihal** — Lecturer

Explore, control, compare

02 INTRODUCTION · A complete project walkthrough, followed by your recorded video



MISSION

Dated launch, four flybys and heliopause.

PROPOSED FEATURES

Earth → interstellar mission
Spacecraft • path • space environment
TRS + hierarchy • spin • cameras
Sun • Flat / Gouraud / Phong • input

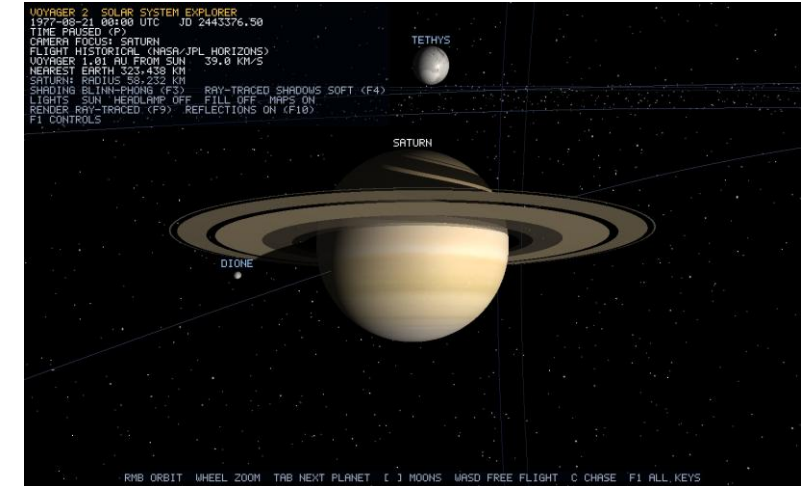


INSPECTION

Body focus, 18 hardware targets, free camera.

IMPLEMENTED + ADDITIONS

26 bodies • 18 hardware targets
Mission restart • manual flight
Extra lights/shaders • surface maps
Shadows • BVH ray tracing



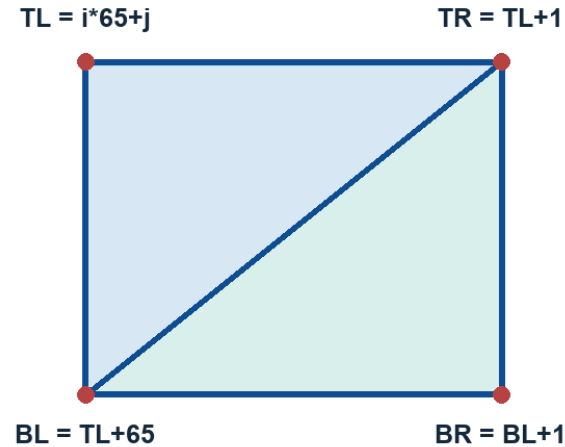
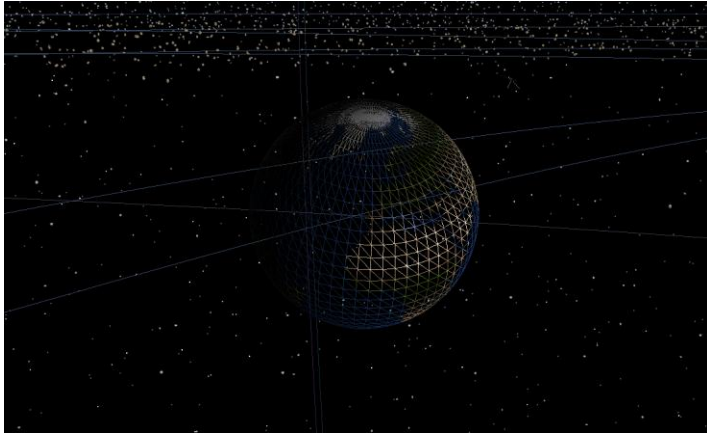
COMPARISON

Three lights, five shading modes, ray tracing.

PLAY PROJECT VIDEO

Vertices → indices → triangles → worlds

03 GEOMETRY · Shared sphere construction creates all 26 textured bodies



Vertex = position + normal + UV

$$\mathbf{p} = (\sin \phi \cos \theta, \cos \phi, \sin \phi \sin \theta)$$

$$\phi = \pi i / L, \quad \theta = 2\pi j / N, \quad uv = (1 - j / N, 1 - i / L)$$

Cell triangles: (TL, TR, BL) and (TR, BR, BL).

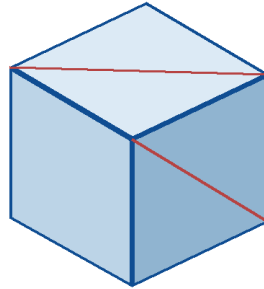
Poles omit degenerate halves; the UV seam duplicates positions.

32 × 64 shared sphere
2,145 vertices · 3,968 triangles
11,904 indices · 26 materials/maps

Voyager: authored from reusable primitives

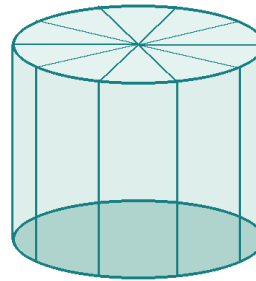
04 OBJECT CONSTRUCTION · Boxes, cylinders, frustums, rods and a curved parabolic dish

BOX / HARD EDGES



24 vertices / 12 triangles

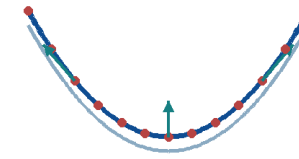
CYLINDER / FRUSTUM



$4n+6$ vertices / $4n$ triangles

PARABOLIC DISH

48 angular segments / 6 radial rings



$$y = -D + D(r/R)^2$$

590 vertices / 1,152 triangles

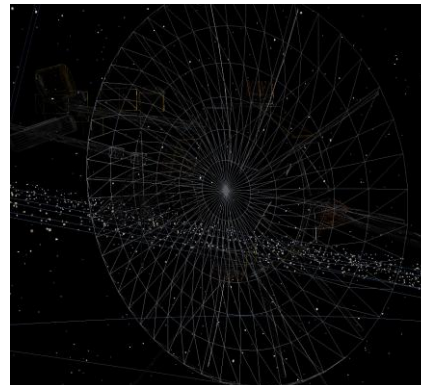
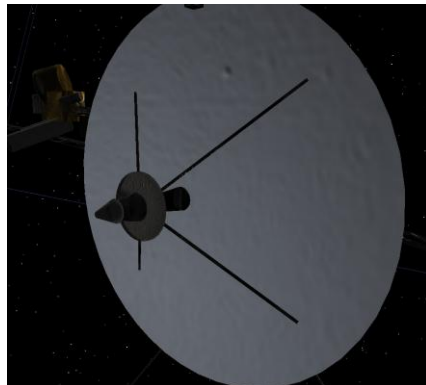
BOX: 24 vertices / 12 triangles

$$y = -D + D(r/R)^2$$

CAPS + SIDES: $4n+6$ / $4n$

$$\mathbf{n} \propto (-2Dx/R^2, 1, -2Dz/R^2)$$

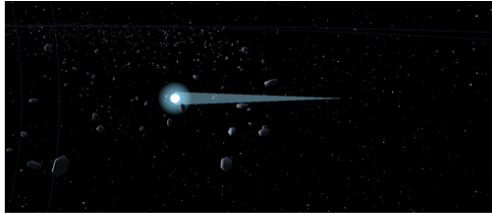
DISH: 590 / 1,152



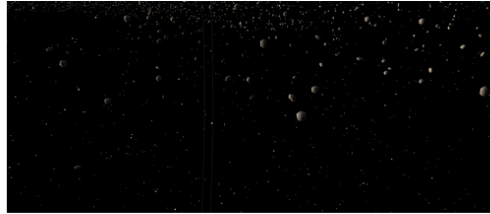
76 assemblies · 5,828 triangles
18 inspectable hardware components
Rods: rotate +Y onto B-A; rebase indices.

Rings, belts, stars, comet and scene guides

05 OBJECT FAMILIES · Every rendered family has a primitive and a construction rule



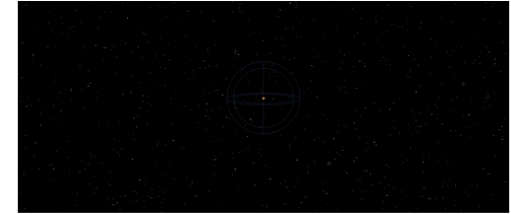
Dynamic comet



Asteroid belt / 4,000



Kuiper belt / 3,000



Oort shell / 1,500

RINGS

$$\mathbf{p} = (r \cos\theta, 0, r \sin\theta)$$

Inner/outer edges; reversed back face.

260 vertices / 256 triangles per band.

ROCK FIELDS

28 vertices / 24 triangles per rock.

8,500 matrices $\rightarrow$ 3 instanced draws.

Deterministic placement; static fields.

STARS / GUIDES

7,380 points; 96-vertex line circles.

No surface triangles for these guides.

HUD text uses screen-space quads.

DYNAMIC COMET = sphere nucleus + glow coma + 16-segment cone tail

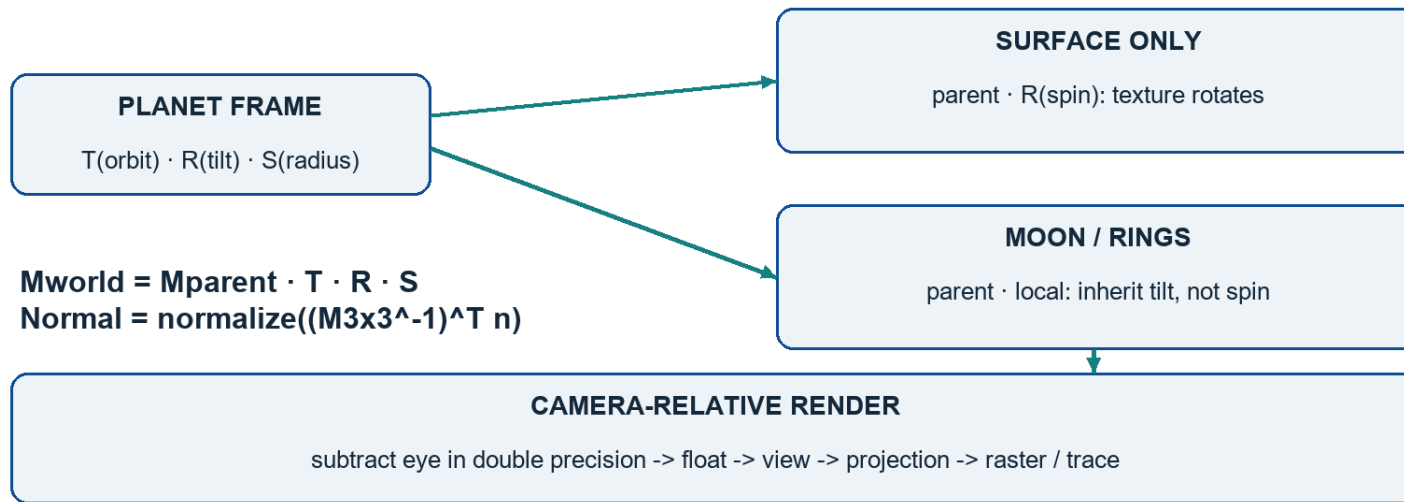
$$\mathbf{d}_{tail} = \text{normalize}(\mathbf{p}_{comet} - \mathbf{p}_{Sun})$$

52 cone vertices / 32 triangles

Transforms, hierarchy, scale and cameras

06 TRANSFORMATIONS · Position, pose, size and view are separate operations

$$\mathbf{p}_{clip} = PVM_{world}\mathbf{p}, \quad M_{world} = M_{parent}TRS$$



$$\mathbf{n}' = \text{normalize}((M_{3 \times 3}^{-1})^T \mathbf{n})$$

Surface-only spin · parent compensation · quaternion rigs · logarithmic depth

Four camera cases

FreeFly: independent motion

Focus: body-centred orbit

Chase: spacecraft frame

Inspect: hardware close-up

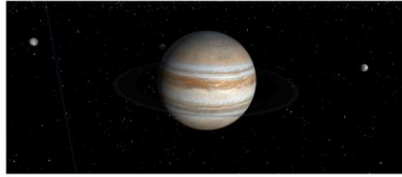
Subtract the eye in double precision.
Compressed scale keeps bodies visible;
physical AU/km values remain in the HUD.

Every motion has an owner and a clock

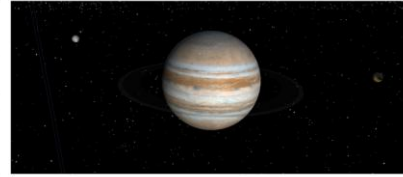
07 MOTION · Dated flight, moon revolution, axial spin, manual six-DOF and comet orientation



Jupiter / initial



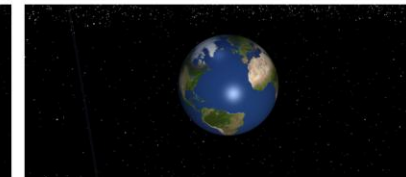
Moons / later



Moons / later still



Spin / initial



Spin / later



Spin / later still

MOON REVOLUTION / tilted parent frame

$$\mathbf{p}(s) = h_{00}\mathbf{p}_0 + h_{10}h\mathbf{v}_0 + h_{01}\mathbf{p}_1 + h_{11}h\mathbf{v}_1$$

$$r = \frac{a(1 - e^2)}{1 + e \cos \nu}, \quad \dot{\nu} = 2\pi(0.4)/P_{days}$$

$$q' = \text{normalize}(qq_y q_x q_z)$$

AXIAL SPIN / surface transform only

$$\dot{\psi} = 2\pi/P_{hours}$$

$$\mathbf{v}' = \mathbf{v} + \mathbf{a}\Delta t, \quad \mathbf{p}' = \mathbf{p} + \mathbf{v}\Delta t$$

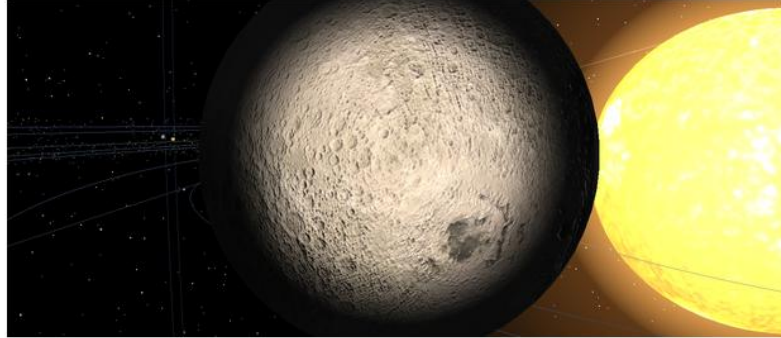
Historical data: one date. Visual moons/spin: independent rates.
Manual flight + cameras: real time. Comet tail: anti-Sun direction.

Three light types—and a moving headlamp

08 LIGHTING · Source geometry and attenuation determine incident illumination



Sun point only



Point + camera spotlight



Point + directional fill

POINT / Sun

$$\mathbf{l} = (\mathbf{p}_L - \mathbf{p}) / d$$

SPOT / camera

$$C = \text{clamp}\left(\frac{\theta - \cos 18^\circ}{\cos 12^\circ - \cos 18^\circ}, 0, 1\right)$$

DIRECTIONAL / fill

$$\mathbf{l} = -\text{normalize}(\mathbf{d}_L)$$

$$A(d) = \frac{1}{k_c + k_l d + k_q d^2}, \quad D = \max(\mathbf{n} \cdot \mathbf{l}, 0)$$

AMBIENT $k_a = 0.07$ | DIFFUSE $D = \max(\mathbf{n} \cdot \mathbf{l}, 0)$
SPECULAR $S_P = k_s \max(\mathbf{r} \cdot \mathbf{v}, 0)^p$
 $I = a(k_a + V_S D_S + D_o) + V_S S_S + S_o$

Camera motion carries the spotlight position and direction; the cone is 12° – 18° .



Frame 1

Frame 2

Frame 3

Five shading techniques, matched views

09 SHADING · Change the normal/evaluation stage while keeping geometry, texture and lights fixed



Flat



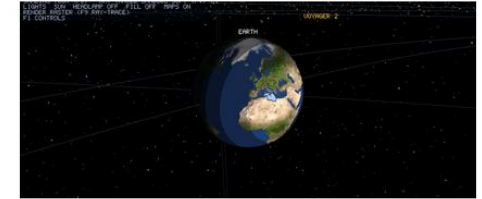
Gouraud



Phong



Blinn-Phong



Toon



Flat



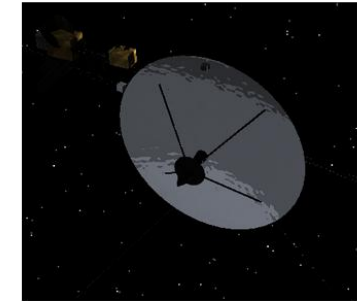
Gouraud



Phong



Blinn-Phong



Toon

FLAT

Face normal

GOURAUD

Vertex light

PHONG

Pixel reflect

BLINN

Pixel half-vector

TOON

Bands + rim

$$S_P = k_s \max(\mathbf{v} \cdot \mathbf{r}, 0)^p$$

$$S_B = k_s \max(\mathbf{n} \cdot \mathbf{h}, 0)^{2p}$$

Toon: 1 / 0.62 / 0.30 / 0.06 bands; rim at $\mathbf{n} \cdot \mathbf{v} < 0.22$. Glow/unlit are separate materials.

Visibility: eclipses, ring shadows and self-shadow

10 SHADOWS · Analytic Sun rays for bodies/rings; a depth map for the spacecraft



Off



Hard



Soft

$$V_{soft} = 1 - \frac{A_{overlap}}{\pi \alpha_{Sun}^2}, \quad V_{ring} = 1 - \text{opacity}$$

BODIES + RINGS

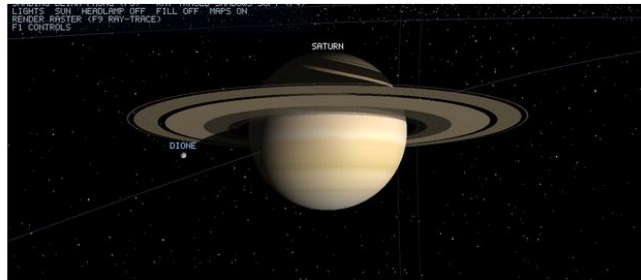
Analytic shadow rays: sphere / annulus.
Finite Sun discs create a stable penumbra.

VOYAGER SELF-SHADOW

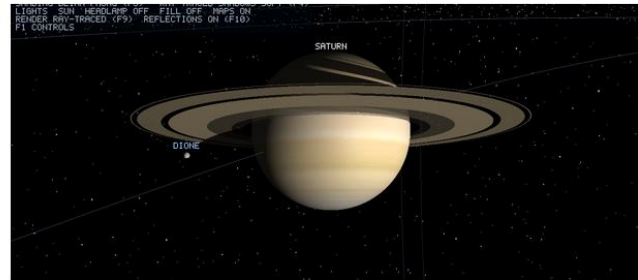
2048² depth map · 3×3 PCF comparisons.
Bias avoids the surface shadowing itself.

Ray tracing: nearest hit, light, reflection

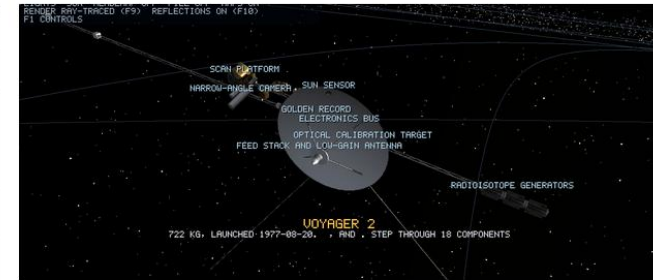
11 RAY TRACING · Analytic spheres/rings and a triangle BVH for the authored spacecraft



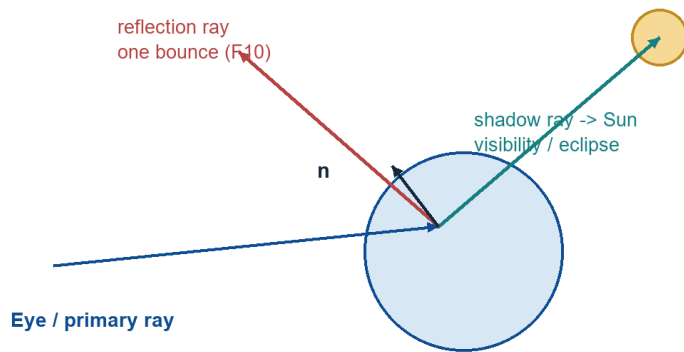
Saturn / raster



Saturn / traced



Voyager / traced BVH



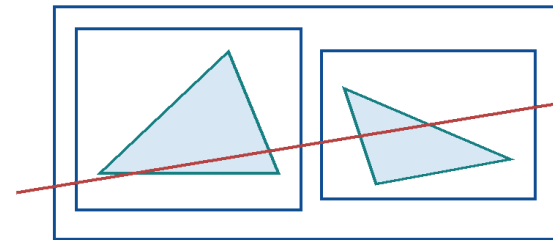
NEAREST HIT

Sphere: quadratic roots
Ring: plane + radius interval
Voyager: triangles in BVH

Then evaluate texture,
normal and Blinn-Phong.
Translucent rings continue
through up to four layers.

Raster view: shadow rays only. F9: primary rays replace body / spacecraft geometry; environment stays rasterized.

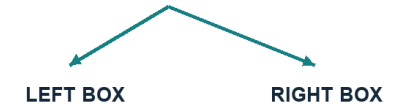
SLAB TEST REJECTS EMPTY BOXES



5,828 triangles / 6,723 nodes / depth 24 / 32-entry traversal stack

CPU builds once. GPU texture buffers store nodes, vertices, normals, UVs and material indices.

ROOT AABB



Near child first; closest t prunes far boxes.
Leaves: up to 4 triangles.
12-bin surface-area heuristic split.

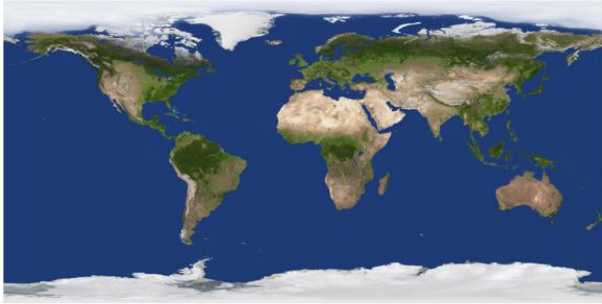
$$\mathbf{p}(t) = \mathbf{o} + t\mathbf{d}, \quad t = -b \pm \sqrt{b^2 - f}$$

$$\mathbf{d}_r = \mathbf{d} - 2(\mathbf{d} \cdot \mathbf{n})\mathbf{n}$$

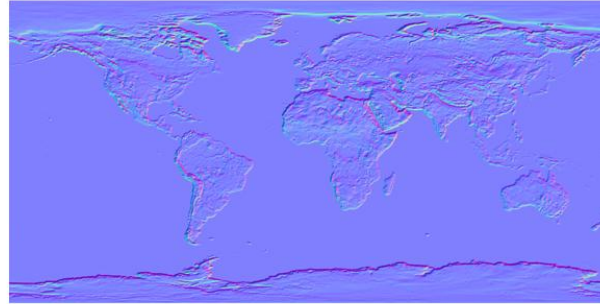
F9 hybrid Whitted view · 5,828 BVH triangles · up to 4 ring layers · F10 one mirror bounce

Textures: colour, atlas, relief and specular mask

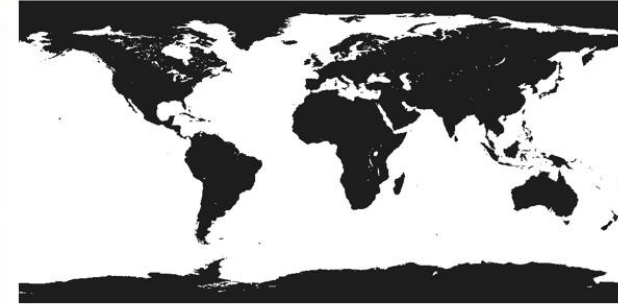
12 TEXTURES · The mesh supplies shape; image channels supply surface detail



Albedo / RGB colour



Sobel-derived normal / approximation



Blue-dominant ocean specular mask



Moon / maps off



Moon / maps on



NASA atlas
11 textured finishes
+ copper material

$$\mathbf{n}_t = \text{normalize}(-sS_x(H), -sS_y(H), 1)$$

$$uv' = uv \cdot zw + xy$$

F8 maps off/on · Sobel relief is inferred from colour · ocean mask controls highlights

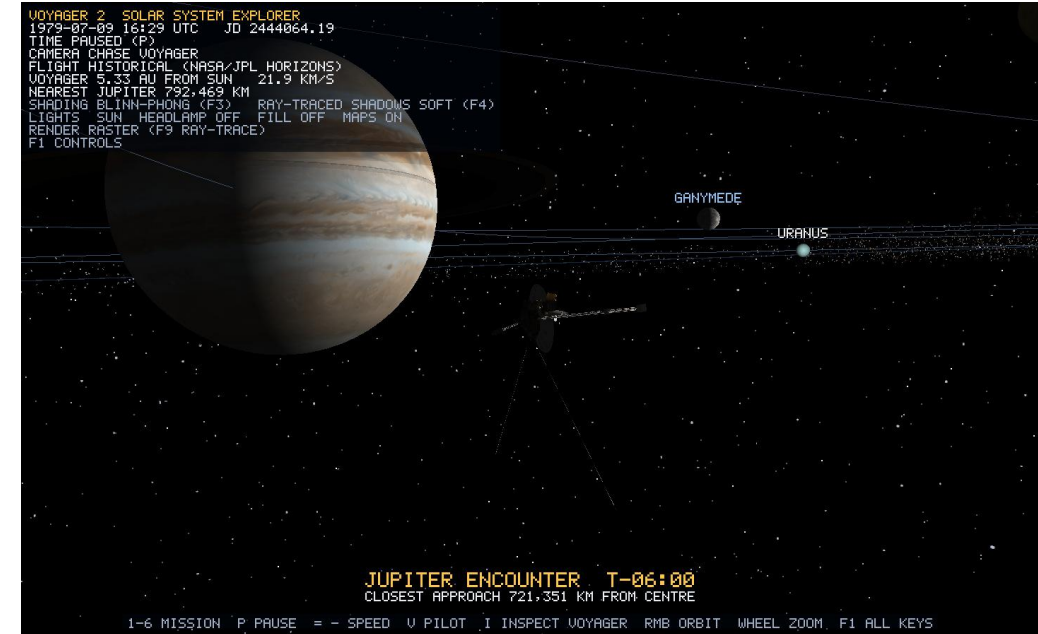
Thank you

13 DISCUSSION · Questions about the project, its mathematics and its implementation

From indexed triangles to an explorable mission.

Geometry · transforms · motion
lighting · shading · ray tracing · textures

Verified: builds, scene checks and runtime captures.
Measured raster views: 0.64–1.15 ms / RX 590.
Explainable limits: compressed scale, visual moons,
hybrid tracing and one reflection bounce.



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